서울시립대 물리학과 고체물리 학부인턴 (2024 여름 비교과프로그램)

대학혁신지원사업

Band Structure of Graphene/h-BN and Stacking Classification in Moiré Patterns Using Machine Learning

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연구 배경 및 목적

This research focuses on understanding stacking area analysis of twisted bilayer materials and modeling machine learning algorithms to classify the obtained stacking data. It also includes band structure analysis using DFT theory and implementation of moiré patterns in twisted bilayer materials.

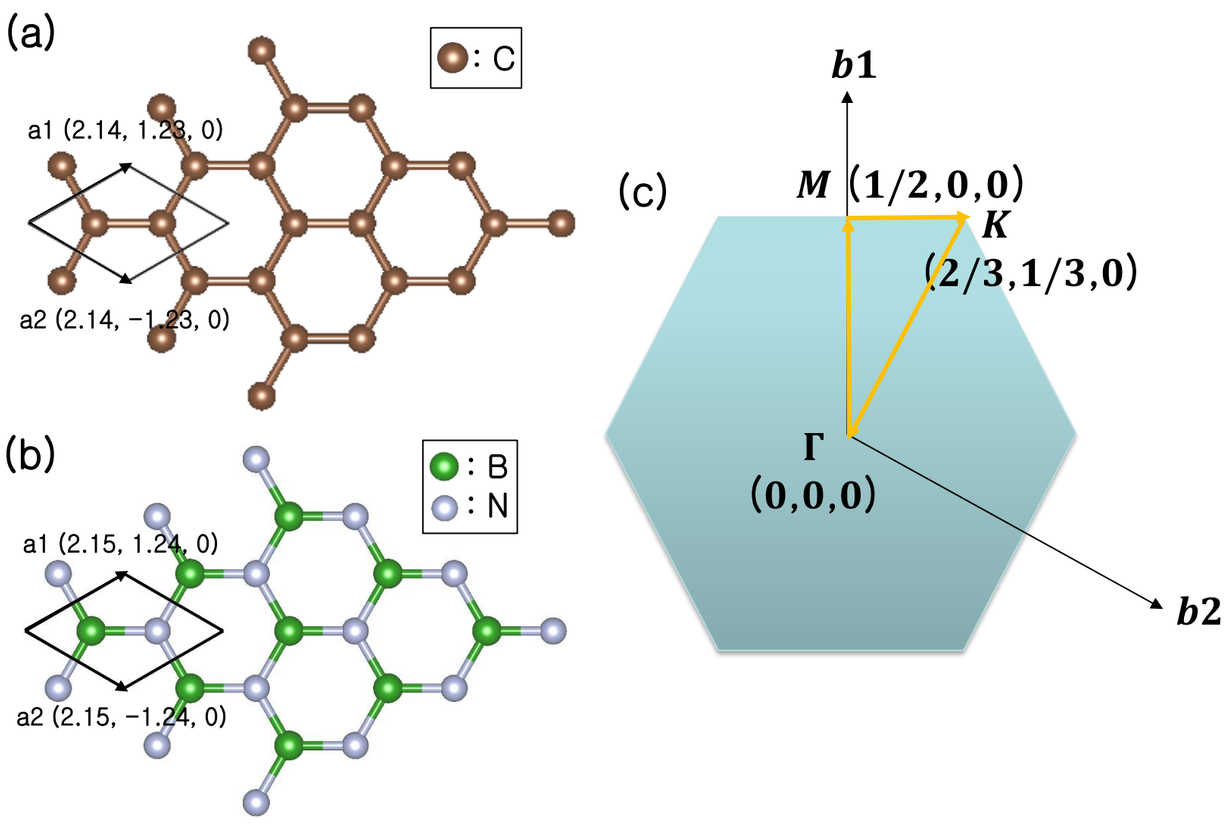
연구 방법

1. Electronic band structure and density of states based-on DFT using QUANTUM ESPRESSO
2. Moiré pattern relaxation using LAMMPS
   Intralayer potential : Extep (Extended Tersoff Potential)
   Interlayer potential : ILP
3. Crysytal visualization : Xcrysden & VESTA & OVITO

연구 결과 및 파급 효과

연구결과

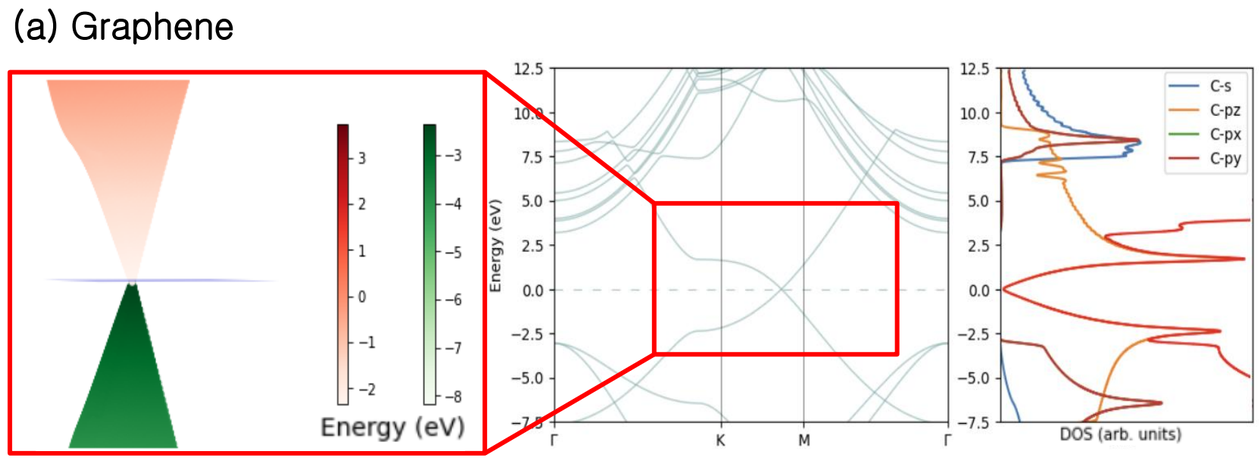
1. Lattice structure & Brillouin zone of graphene and h_BN



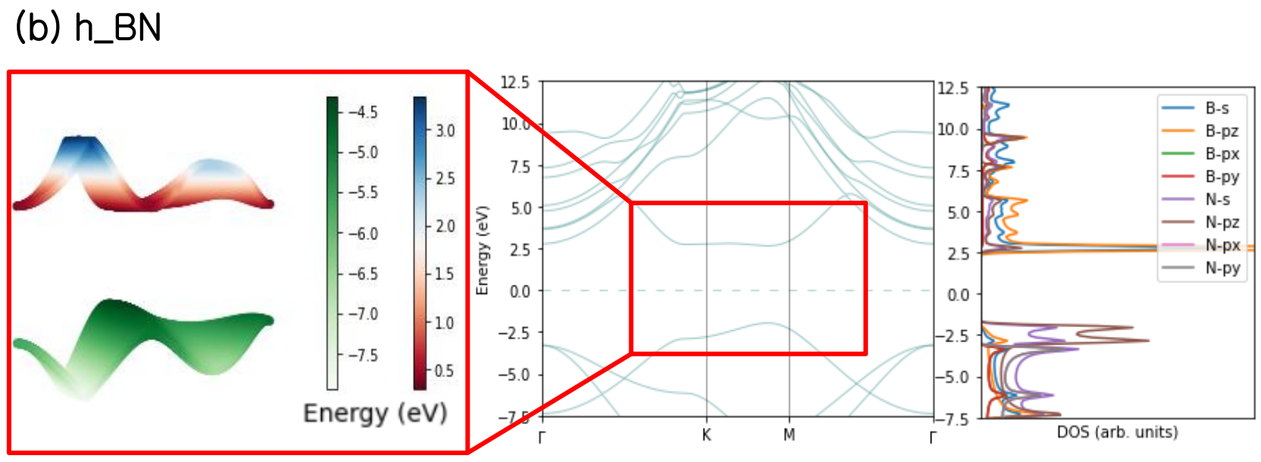
(a) : Honeycomb lattice of graphene, a1 and a2 are the lattice vector
(b) : Honeycomb lattice of h_BN, a1 and a2 are the lattice vector
(c) : First Brillouin zone of graphene and h_BN, b1 and b2 are the reciprocal lattice vectors.

2.Band Structure & Density Of State

(a) Graphene

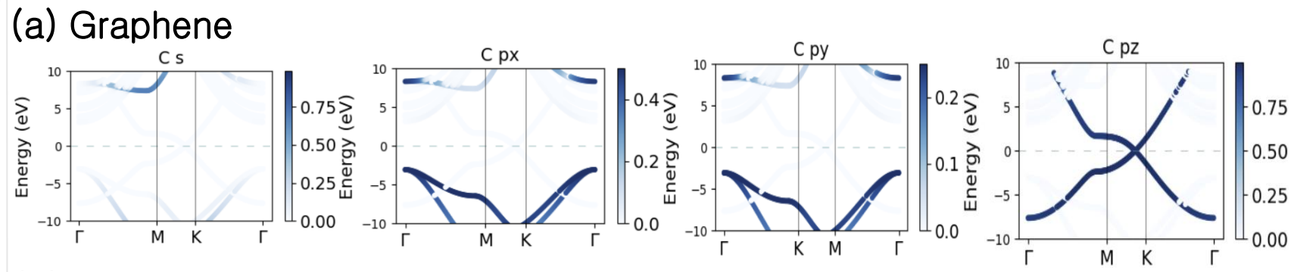


(b) h_BN

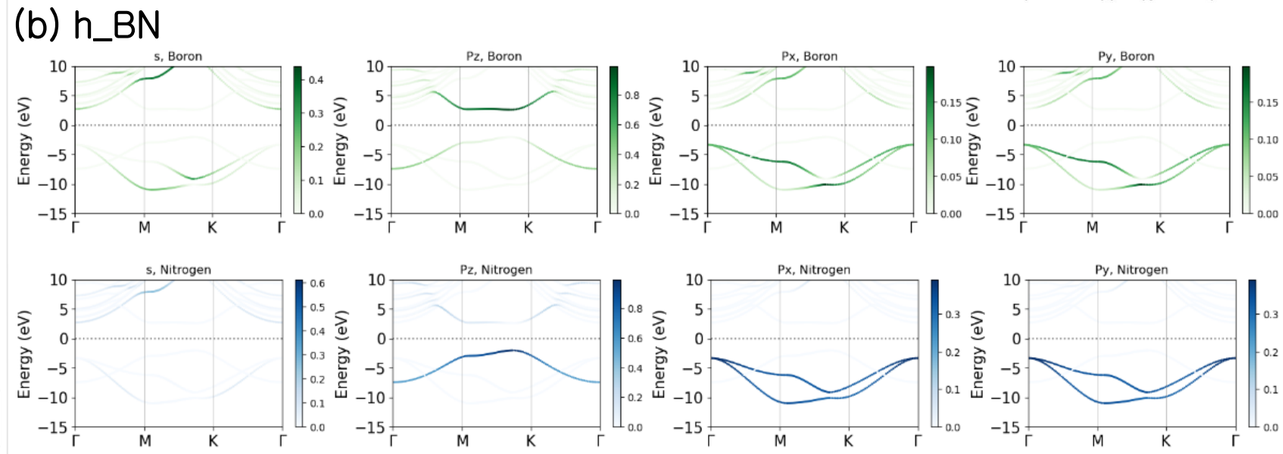


3.Projected Band Structure

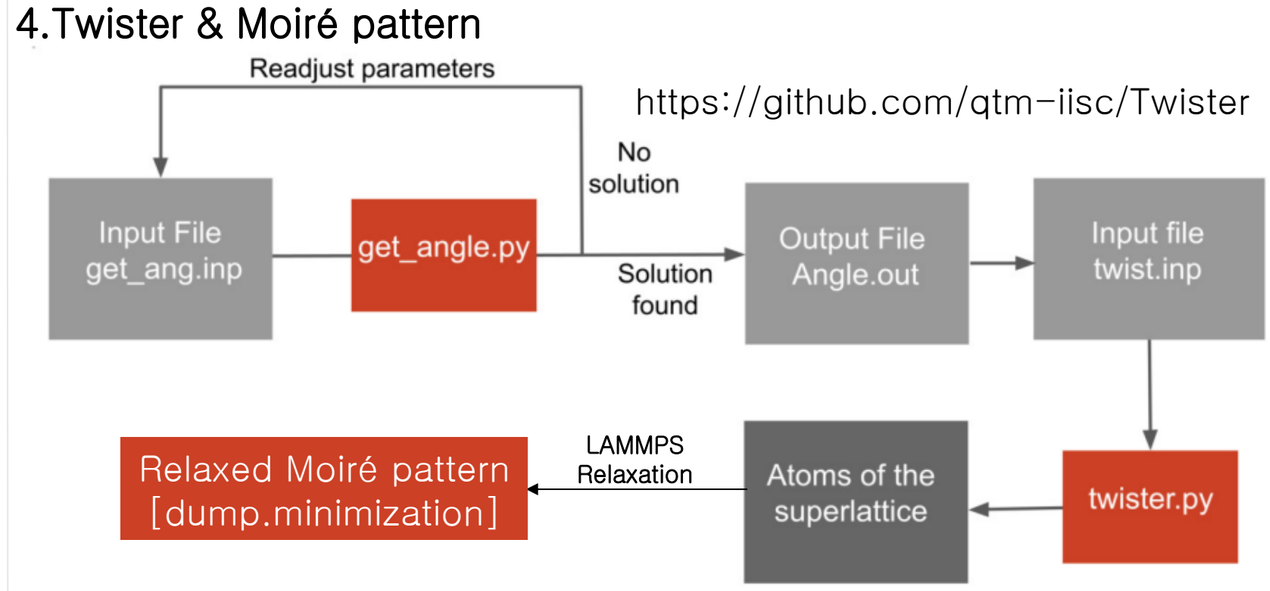
(a) Graphene



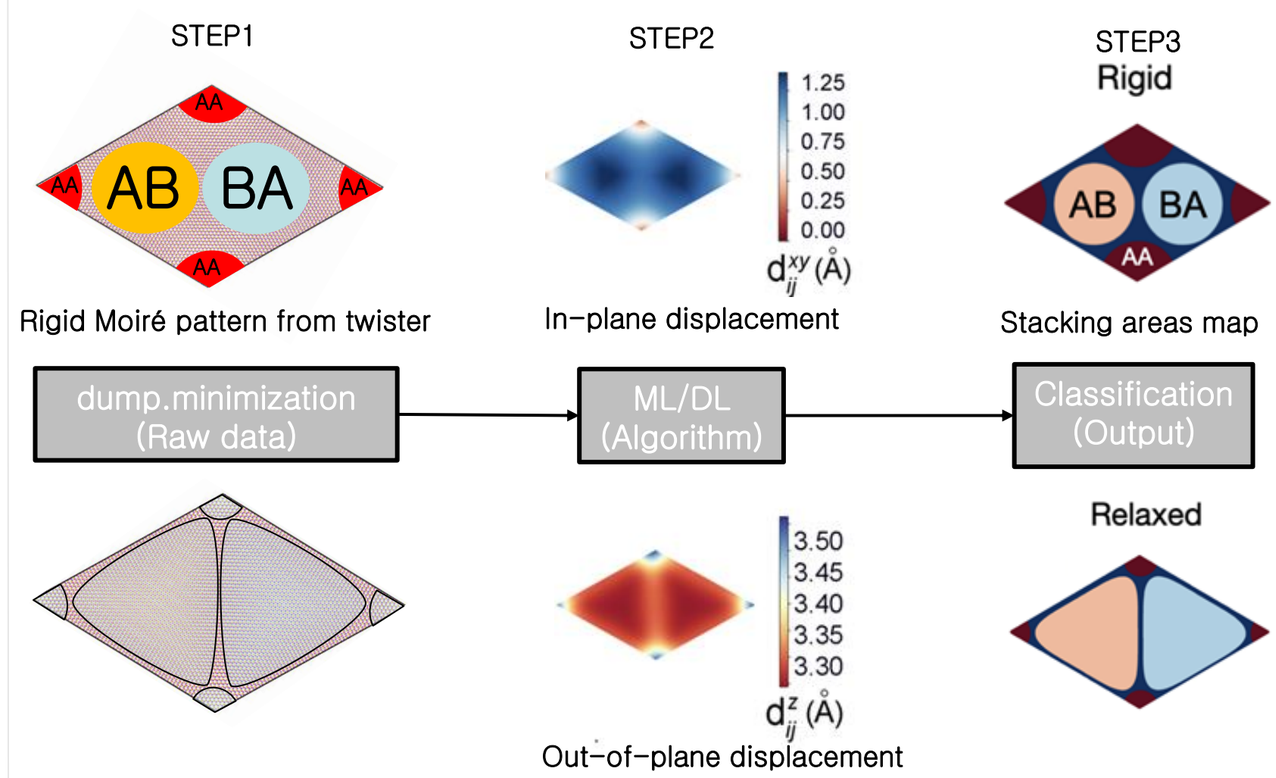
(b) h_BN



4.Twister & Moiré pattern



5.Machine Learning (Twisted bilayer h_BN 1.08°)



연구결과의 파급효과

This research aims to understand the principles of implementing stacking regions in 2D materials using DFT and to develop machine learning algorithms for efficient image classification. As stacking images of 2D materials become increasingly diverse, effective machine learning methods will greatly enhance research efficiency.

참조문헌

1. Li, F., Lee, D., Leconte, N., Javvaji, S., & Jung, J. (2024), 「Moiré flat bands and antiferroelectric domains in lattice relaxed twisted bilayer hexagonal boron nitride under perpendicular electric fields」,arXiv preprint arXiv:2406.12231.
2. Naik, S., Naik, M. H., Mairy, I., & Jain, M. (2022). 「Twister: Construction and structural relaxation of commensurate Moiré superlattices」, ScienceDirect.